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Patent Public Search | Text View

United States Patent Application Publication

20250274123

Kind Code

A1

Publication Date

August 28, 2025

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COMPACT HIGH VOLTAGE PRE-DRIVER AND CIRCUITS INCLUDING THE PRE-DRIVER

Abstract

A pre-driver includes first, second, third, and fourth P-channel field effect transistors (PFETs) with voltage ratings equal to a low first voltage (V1). Between power rails at V1 and at a higher second voltage (V2), the first and third PFETs are series-connected and the second and fourth PFETs are also series-connected. The third and fourth PFETs are cross-coupled. The first and second PFETs have gates that receive a pre-driver input signal (Pin) that switches between V2 and ground and an inverted pre-driver input signal (Pinb), respectively. At an output node between the first and third PFETs, the pre-driver outputs a driver input signal (Din) that switches between V2 and V1. A circuit includes this pre-driver (e.g., connected between a voltage level shifter and a driver or connected between a voltage level shifter and a buffer).

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Family ID: 1000007712425

Appl. No.: 18/585187

Filed: February 23, 2024

Publication Classification

Int. Cl.: H03K19/0185 (20060101); H03K17/687 (20060101)

U.S. Cl.:

CPC H03K19/018507 (20130101); H03K17/6871 (20130101);

Background/Summary

BACKGROUND

[0001] The present disclosure relates to pre-drivers and, more particularly, to embodiments of a compact high voltage pre-driver and to circuits that can include the compact high voltage pre-driver.

[0002] Memory structures typically include, among other components: an array of memory cells arranged in rows and columns; wordlines (WLs) for the rows; and wordline driver circuits for the WLs. A WL for a row is connected to all memory cells in that row and also connected to a wordline driver circuit. The wordline driver circuit receives a first wordline select signal (e.g., from a row address decoder) and outputs a second wordline select signal to the WL to provide access to selected memory cell(s) connected to the WL during read/write operations. In some memory structures, the first wordline select signal received by the wordline driver circuit may be a relatively low voltage wordline select signal (WLsel-lv) (e.g., for reduced power consumption). However, for improved performance during read/write operations, it may be advantageous for the second wordline select signal output to the WL itself to be a relatively high voltage wordline select signal (WLsel-hv). Therefore, multi-stage wordline driver circuits have been developed that provide low voltage to high voltage signal conversion. These multi-stage wordline driver circuits typically include three stages. A first stage (also referred to herein as a voltage level shifter) converts a received WLsel-lv, which switches between a relatively low positive first voltage (V1) and ground (GND), to a pre-driver input signal (Pin), which switches between a relatively high positive second voltage (V2) and GND. A second stage (also referred to herein as a pre-driver) converts Pin to a driver input signal (Din), which switches between V2 and V1. A third stage (also referred to herein as a driver) converts Din to WLsel-hv, which switches between V2 and GND and which is output to the WL. However, with such multi-stage wordline driver circuits, there are often undesirable trade-offs between area, power, and performance.

SUMMARY

[0003] Disclosed herein are embodiments of a pre-driver. The pre-driver can include a first input node and a second input node. The pre-driver can further include four transistors and, particularly, a first P-channel field effect transistor (PFET) with a first gate connected to the first input node, a second PFET with a second gate connected to the second input node, a third PFET, and a fourth PFET. The first PFET and the third PFET can be connected in series between a first power rail at a first voltage and a second power rail at a second voltage that is greater than (e.g., twice) the first voltage. The second PFET and the fourth PFET can also be connected in series between the first power rail and the second power rail. Furthermore, the third PFET and the fourth PFET can be cross-coupled. In this pre-driver the first input node can be connected to receive a pre-driver input signal that switches between the second voltage and ground and the second input node can be connected to receive an inverted pre-driver input signal. The pre-driver can also include an output node at a junction between the first PFET and the third PFET and the output node can output a pre-driver output signal that switches between the second voltage and the first voltage.

[0004] Also disclosed herein are embodiments of circuits that include such a pre-driver. For example, a disclosed circuit can include a voltage level shifter, which is connected to receive a voltage level shifter input signal that switches between a first voltage and ground. The voltage level shifter can further output a pre-driver input signal that switches between a second voltage that is greater than (e.g., twice) the first voltage and ground and an inverted pre-driver input signal. The circuit can further include a pre-driver. The pre-driver can include a first input node and a second input node. The pre-driver can further include four transistors and, particularly, a first P-channel field effect transistor (PFET) with a first gate connected to the first input node, a second PFET with

a second gate connected to the second input node, a third PFET, and a fourth PFET. The first PFET and the third PFET can be connected in series between a first power rail at a first voltage and a second power rail at a second voltage that is greater than (e.g., twice) the first voltage. The second PFET and the fourth PFET can also be connected in series between the first power rail and the second power rail. Furthermore, the third PFET and the fourth PFET can be cross-coupled. In this pre-driver the first input node can be connected to receive the pre-driver input signal and the second input node can be connected to receive the inverted pre-driver input signal. The pre-driver can also include an output node at a junction between the first PFET and the third PFET and the output node can output a pre-driver output signal that switches between the second voltage and the first voltage. The circuit can further include a device, which is connected to receive the pre-driver output signal. The device can be, for example, a driver, a buffer, etc.

[0005] One embodiment of a circuit disclosed herein can specifically be a multi-stage wordline driver circuit. This circuit can include a voltage level shifter, which is connected to receive a first wordline select signal that switches between a first voltage and ground. The voltage level shifter can further output a pre-driver input signal that switches between a second voltage that is greater than (e.g., twice) the first voltage and ground and an inverted pre-driver input signal. The circuit can further include a pre-driver. The pre-driver can include a first input node and a second input node. The pre-driver can further include four transistors and, particularly, a first P-channel field effect transistor (PFET) with a first gate connected to the first input node, a second PFET with a second gate connected to the second input node, a third PFET, and a fourth PFET. The first PFET and the third PFET can be connected in series between a first power rail at a first voltage and a second power rail at a second voltage that is greater than (e.g., twice) the first voltage. The second PFET and the fourth PFET can also be connected in series between the first power rail and the second power rail. Furthermore, the third PFET and the fourth PFET can be cross-coupled. In this pre-driver the first input node can be connected to receive the pre-driver input signal and the second input node can be connected to receive the inverted pre-driver input signal. The pre-driver can also include an output node at a junction between the first PFET and the third PFET and the output node can output a pre-driver output signal (also referred to herein as a driver input signal) that switches between the second voltage and the first voltage. The circuit can further include a driver, which is connected to receive the pre-driver output signal (i.e., the driver input signal) and which outputs a second wordline select signal that switches between the second voltage and ground.

[0006] It should be noted that all aspects, examples, and features of disclosed embodiments mentioned in the summary above can be combined in any technically possible way. That is, two or more aspects of any of the disclosed embodiments, including those described in this summary section, may be combined to form implementations not specifically described herein. The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure will be better understood from the following detailed description with reference to the drawings, which are not necessarily drawn to scale and in which:

[0008] FIG. 1 is a schematic diagram illustrating a disclosed embodiment of a multi-stage wordline driver circuit;

[0009] FIG. 2A is a schematic diagram illustrating a disclosed embodiment of a pre-driver that can be incorporated into the circuit of FIG. 1;

[0010] FIG. 2B is a cross-section diagram illustrating an example of the pre-driver of FIG. 2A implemented as a semiconductor-on-insulator structure;

[0011] FIG. 3 is a schematic diagram illustrating an example of a voltage level shifter that can, for example, be incorporated into the circuit of FIG. 1;

[0012] FIG. 4 is a schematic diagram illustrating an example of a driver that can, for example, be incorporated into the circuit of FIG. 1;

[0013] FIG. 5 is a schematic diagram illustrating a disclosed embodiment of a multi-stage variable power supply circuit;

[0014] FIG. 6 is a schematic diagram illustrating a disclosed embodiment of a pre-driver that can be incorporated into the circuit of FIG. 5; and

[0015] FIG. 7 is a graph illustrating pre-driver performance in an example implementation.

[0016] It should be noted that all aspects, examples, and features of disclosed embodiments mentioned in the summary above can be combined in any technically possible way. That is, two or more aspects of any of the disclosed embodiments, including those described in this summary section, may be combined to form implementations not specifically described herein. The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects and advantages will be apparent from the description and drawings, and from the claims.

DETAILED DESCRIPTION

[0017] As mentioned above, in some memory structures, a first wordline select signal received by a wordline driver circuit may be a relatively low voltage wordline select signal (WLsel-lv) (e.g., for reduced power consumption). However, for improved performance during read/write operations, it may be advantageous for the second wordline select signal output to the WL itself to be a relatively high voltage wordline select signal (WLsel-hv). Therefore, multi-stage wordline driver circuits have been developed that provide low voltage to high voltage signal conversion. These multi-stage wordline driver circuits typically include three stages. A first stage (also referred to herein as a voltage level shifter) converts a received WLsel-lv, which switches between a relatively low positive first voltage (V1) and ground (GND), to a pre-driver input signal (Pin), which switches between a relatively high positive second voltage (V2) and GND. A second stage (also referred to herein as a pre-driver) converts Pin to a driver input signal (Din), which switches between V2 and V1. A third stage (also referred to herein as a driver) converts Din to WLsel-hv, which switches between V2 and GND and which is output to the WL. However, with such multi-stage wordline driver circuits, there are often undesirable trade-offs between area, power, and performance.

[0018] For example, as mentioned above, the pre-driver of a multi-stage wordline driver circuit converts Pin to Din. To accomplish this, a conventional pre-driver includes a pair of series-connected inverters and, thus, a combination of both N-channel field effect transistors (NFETs) and P-channel field effect transistors (PFETs). The use of both NFETs and PFETs requires a significant amount of area (e.g., due to bulk spacing requirements) and also results in significant process, voltage, and temperature (PVT) variations. Furthermore, the NFETs and PFETs used include either: (a) all low voltage devices with at least some of the low voltage devices necessarily operating outside the safe operating area (SOA) (e.g., above the voltage ratings for the operating drain-source voltage (VDS), the operating gate-source voltage (VGS), or the operating gate-drain voltage (VGD)) or (b) a combination of both low voltage devices and high voltage devices with the high voltage devices being, for example, relatively large laterally double-diffused metal oxide semiconductor field effect transistors (LDMOSFETs) that have a slower switching speed.

[0019] In view of the foregoing, disclosed herein are embodiments of a novel compact high voltage pre-driver. The pre-driver can include four P-channel field effect transistors (PFETs) and, particularly, a first PFET, a second PFET, a third PFET and a fourth PFET. These PFETs can be low voltage PFETs, all with identical voltage ratings for VDS, VGS, and VGD. The first and third PFETs can be series-connected between a first power rail set at a relatively low first voltage (V1)

and a second power rail at a relatively high second voltage (V2) (e.g., $V2 \sim 2 \cdot V1$). V1 can, for example, be set to correspond to the voltage rating for VDS, VGS, and VGD. The second and fourth PFETs can be series-connected between these two power rails. The third and fourth PFETs can be cross-coupled. The first and second PFETs can have gates connected to receive a pre-driver input signal (Pin) that switches between V2 and ground (GND) and an inverted pre-driver input signal (Pinb). The pre-driver can further include an output node at the junction between the first and third PFETs and this output node can output a pre-driver output signal (Pout) (also referred to herein as a driver input signal (Din)). Din can switch between V2 and V1. In the disclosed pre-driver, using only PFETs reduces area consumption because bulk spacing requirements between PFETs and NFETs are avoided. Furthermore, in the disclosed pre-driver, none of the low voltage PFETs operate outside the SOA (i.e., voltage ratings are not violated). Finally, the disclosed pre-driver exhibits improved performance (e.g., reduced delay, reduced static and short circuit currents, etc.) and reduced PVT variations. Also disclosed herein are embodiments of multi-stage circuits that can include the disclosed pre-driver. For example, a multi-stage wordline driver circuit can include the pre-driver between a voltage level shifter and a driver. In this case, the voltage level shifter can receive a low voltage wordline select signal (WLsel-lv), which switches between V1 and ground, and can output Pin and Pinb to the pre-driver. As mentioned above, the pre-driver can receive Pin and Pinb and output Din. The driver can receive Din and, in response, can output a high voltage wordline select signal (WLsel-hv) to a wordline and this WLsel-hv can switch between V2 and ground.

[0020] FIG. 1 is a schematic diagram illustrating a multi-stage wordline driver circuit **100**. Multi-stage wordline driver circuit **100** can include a voltage level shifter **110**, a driver **130**, and a novel compact high voltage pre-driver **120**, which is connected to voltage level shifter **110** and to driver **130**. In some embodiments, the multi-stage wordline driver circuit **100** and components thereof, including the compact high voltage pre-driver **120**, can be implemented in an advanced semiconductor-on-insulator processing technology node (e.g., a fully depleted semiconductor-on-insulator processing technology node, such as a fully depleted silicon-on-insulator (FDSOI) processing technology node).

[0021] FIG. 2A is schematic diagram illustrating, in greater detail, a disclosed embodiment of a compact high voltage pre-driver **120** (hereinafter referred to simply as pre-driver **120**) that can, for example, be incorporated into the multi-stage wordline driver circuit **100** of FIG. 1. FIG. 2B is a cross-section diagram illustrating an example of pre-driver **120** implemented as semiconductor-on-insulator structure.

[0022] A semiconductor-on-insulator structure can include a semiconductor substrate **201**. Semiconductor substrate **201** can be, for example, a monocrystalline silicon substrate or, alternatively, a monocrystalline substrate of any other suitable semiconductor material (e.g., silicon germanium, etc.). The semiconductor-on-insulator structure can further include an insulator layer **205** on semiconductor substrate **201**. Insulator layer **205** can be, for example, a silicon dioxide layer or a layer of any other suitable insulator material. The semiconductor-on-insulator structure can further include a semiconductor layer **206** on insulator layer **205**. The semiconductor layer **206** can be, for example, a monocrystalline silicon layer or a layer of any other suitable monocrystalline semiconductor material (e.g., silicon germanium, etc.).

[0023] Pre-driver **120** can be electrically connected between a first power rail **291** at a relatively low first voltage (V1) and a second power rail **292** at a relatively high second voltage (V2). V1 and V2 can both be positive voltages. Additionally, V2 can be higher than V1 and, more specifically, can be essentially double the magnitude of V1. For purposes of this disclosure, “essentially double the magnitude” refers to twice the magnitude, plus or minus 10% thereof (e.g., if V1 is 1.5V, then $V2 = 3.0V \pm 0.3V$; if V1 is 1.8V, then $V2 = 3.6V \pm 0.36V$; and so on).

[0024] Pre-driver **120** can further include four PFETs **210**, **220**, **230**, **240** and, particularly, a first PFET **210**, a second PFET **220**, a third PFET **230** and a fourth PFET **240**. First PFET **210** can

include a first source region **211**, a first drain region **212**, a first channel region **215** between first source region **211** and first drain region **212**, and a first gate **213** adjacent to first channel region **215**. Second PFET **220** can include a second source region **221**, a second drain region **222**, a second channel region **225** between second source region **221** and second drain region **222**, and a second gate **223** adjacent to second channel region **225**. Third PFET **230** can include a third source region **231**, a third drain region **232**, a third channel region **235** between third source region **231** and third drain region **232**, and a third gate **233** adjacent to third channel region **235**. Fourth PFET **240** can include a fourth source region **241**, a fourth drain region **242**, a fourth channel region **245** between fourth source region **241** and fourth drain region **242**, and a fourth gate **243** adjacent to fourth channel region **245**.

[0025] Each PFET **210**, **220**, **230**, and **240** can be a low voltage symmetric PFET. Additionally, PFETs **210**, **220**, **230**, and **240** can have the same relatively low voltage ratings for the maximum operating VDS, the maximum operating VGS, and the maximum operating VGD. V1 can be set so that it corresponds to this relatively low voltage rating. For example, if the voltage ratings for maximum operating VDS, maximum operating VGS, and maximum operating VGD of PFETs **210**, **220**, **230** and **240** are all 1.8V, then V1 can also be 1.8V. In this case, V2 would be 3.6V. If the voltage ratings for maximum operating VDS, maximum operating VGS, and maximum operating VGD of PFETs **210**, **220**, **230** and **240** are all 1.5V, then V1 can also be 1.5V. In this case, V2 would be 3.0V, and so on. In some embodiments, first PFET **210** and second PFET **220** can have a first size and third PFET **230** and fourth PFET **240** can have second size that is different from the first size (e.g., smaller than the first size).

[0026] First PFET **210** and third PFET **230** can be series-connected between first power rail **291** and second power rail **292**. That is, in first PFET **210**, first drain region **212** can be electrically connected to first power rail **291** and first source region **211** can be electrically connected to third drain region **232** of third PFET **230**. Additionally, in third PFET **230**, third source region **231** can be electrically connected to second power rail **292**. Similarly, second PFET **220** and fourth PFET **240** can be series-connected between first power rail **291** and second power rail **292**. That is, in second PFET **220**, second drain region **222** can be electrically connected to first power rail **291** and second source region **221** can be electrically connected to fourth drain region **242** of fourth PFET **240**. Additionally, in fourth PFET **240**, fourth source region **241** can be electrically connected to second power rail **292**. Thus, the series-connected first and third PFETs are connected in parallel with the series-connected second and fourth PFETs. Additionally, third PFET **230** and fourth PFET **240** can be cross-coupled. That is, third gate **233** of third PFET **230** can be electrically connected to fourth drain region **242** of fourth PFET **240** and fourth gate **243** of fourth PFET **240** can be electrically connected to third drain region **232** of third PFET **230**.

[0027] Pre-driver **120** can further include a first pre-driver input node **251** electrically connected to first gate **213** of first PFET **210** and a second pre-driver input node **252** electrically connected to second gate **223** of second PFET **220**. First pre-driver input node **251** can be electrically connected (e.g., to one output node of a voltage level shifter **110** of a multi-stage wordline driver circuit **100**, as discussed in greater detail below) to receive a pre-driver input signal (Pin) **118**. Pin **118** can switch between V2 and ground (GND) (e.g., 0.0V). Second pre-driver input node **252** can be electrically connected (e.g., to another output node of voltage level shifter **110**, as discussed in greater detail below) to receive an inverted pre-driver input signal (Pinb) **119**. Pinb **119** can be inverted with respect to Pin **118** such that, when Pin **118** switches from V2 to GND, Pinb **119** switches from GND to V2 and vice versa.

[0028] Pre-driver **120** can further include a first pre-driver output node **261** and a second pre-driver output node **262**. First pre-driver output node **261** can be electrically connected to the junction between first source region **211** of first PFET **210** and third drain region **232** of third PFET **230** (and, thereby to fourth gate **243** of fourth PFET **240**). First pre-driver output node **261** can output a pre-driver output signal (Pout) (also referred to in pre-driver **120** as a driver input signal (Din)

128). **Din 128** can switch between **V2** and **V1**. Second pre-driver output node **262** can be electrically connected to the junction between second source region **221** of second PFET **220** and fourth drain region **242** of fourth PFET **240** (and, thereby to third gate **233** of third PFET **230**). Second pre-driver output node **262** can output an inverted pre-driver output signal (**Poutd**) (also referred to in pre-driver **120** as an inverted driver input signal (**Dinb**) **129**). **Dinb 129** can inverted with respect to **Din 128** such that, when **Din 128** switches from **V2** to **V1**, **Dinb 129** switches from **V1** to **V2** and vice versa.

[0029] As illustrated in FIG. 2B, in some embodiments, channel regions **215**, **225**, **235**, **245** of PFETs **210**, **220**, **230**, **240** can be within semiconductor layer **206** and positioned laterally between corresponding source/drain regions **211-212**, **221-222**, **231-232**, **241-242**, respectively. Source/drain regions **211-212**, **221-222**, **231-232**, **241-242** can have P-type conductivity at a relatively high conductivity level) (e.g., can be P+ source/drain regions). Channel regions **215**, **225**, **235**, **245** can either have N-type conductivity at a relatively low conductivity level or can be undoped (e.g., can be N- or intrinsic channel regions). As illustrated, some of the PFETs can have shared source/drain regions. For example, first source region **211** of first PFET **210** and third drain region **232** of third PFET **230** (which are series-connected and tied to first pre-driver output node **261**) can be shared (e.g., within the same doped region of semiconductor layer **206**). Additionally, second source region **221** of second PFET **220** and fourth drain region **242** of fourth PFET **240** (which are series-connected and tied to second pre-driver output node **262**) can be shared. Finally, third source region **231** of third PFET **230** and fourth source region **241** of fourth PFET **240** (which connected in parallel to second power rail **292**) can be shared.

[0030] As illustrated in FIG. 2B, each of the source/drain regions **211-212**, **221-222**, **231-232**, **241-242** of PFETs **210**, **220**, **230**, **240** can include a lower portion within semiconductor layer **206** and, optionally, an upper portion (e.g., a situ doped epitaxial semiconductor layer) on the top surface of semiconductor layer **206**. Gates **213**, **223**, **233**, **243** of PFETs **210**, **220**, **230**, **240** can be above semiconductor layer **206** immediately adjacent to channel regions **215**, **225**, **235**, **245**, respectively. In some embodiments, gates **213**, **223**, **233**, **243** can be gate-first high-K metal gate (HKMG) structures. Those skilled in the art will recognize that a gate-first HKMG structure can include, for example, a gate dielectric layer that includes: an interfacial layer (e.g., an SiO₂ layer and/or an SiON layer) and a high-K gate dielectric layer on the interfacial layer. The high-K gate dielectric layer can be a layer of dielectric material with a dielectric constant that is greater than 3.9 including, for example, hafnium (Hf)-based dielectrics, such as hafnium oxide, hafnium silicon oxide, hafnium silicon oxynitride, or hafnium aluminum oxide, or other suitable high-k dielectrics, such as aluminum oxide, tantalum oxide, or zirconium oxide.

[0031] The gate-first HKMG structure can further include, for example, a gate conductor layer that includes at least: one or more stacked metal layers (e.g., a metal capping layer and an additional metal material layer suitable for dipole formation on the metal capping layer) on the high-K gate dielectric layer and an N-doped polysilicon gate conductor layer on the metal gate conductor layer(s). Alternatively, gates **213**, **223**, **233**, **243** of PFETs **210**, **220**, **230**, **240** can be any other suitable type of front gate structure (e.g., a gate first polysilicon gate structure, a replacement metal gate structure, etc.) used, for example, in fully depleted semiconductor-on-insulator (e.g., in FDSOI) processing technology platforms. In any case, gate sidewall spacers can be positioned laterally adjacent to opposing sidewalls of gates **213**, **223**, **233**, **243** and can electrically isolate the gates from adjacent source/drain regions. Gate sidewall spacers are well known in the art and, thus, the details thereof have been omitted from this specification in order to allow the reader to focus on the salient aspects of the disclosed embodiments.

[0032] Those skilled in the art will recognize that one advantage of fully depleted semiconductor-on-insulator (e.g., FDSOI) technology processing platforms is that FETs formed above insulator layer above either a well region (e.g., an Nwell or a Pwell) in order to achieve different types of NFETs or PFETs with different VTs. For purposes of this disclosure, a well region refers to a doped

region of a semiconductor material. A Pwell is doped so as to have P-type conductivity, whereas an Nwell is doped so as to have N-type conductivity. For super low threshold voltage (SLVT) or low threshold voltage (LVT) devices, PFETs can be on an insulator layer aligned above a Pwell. For regular threshold voltage (RVT) or high threshold voltage (HVT) devices, PFETs can be on an insulator layer aligned above an Nwell. Whether PFETs on an insulator layer over a Pwell are SLVT or LVT devices or whether PFETs on an insulator layer over an Nwell are RVT or HVT devices will depend upon the design (e.g., device size, etc.) and process specifications (e.g., dopant concentrations, etc.). Another advantage of fully depleted semiconductor-on-insulator (e.g., FDSOI) processing technology platforms is back biasing. Specifically, in such structures, adjacent portions of the insulator layer and a well region aligned below the channel region of an FDSOI FET can effectively form a back gate. The back gate can be biased by biasing the well region to fine tune the V_T . Forward back biasing (FBB) refers to back biasing that reduces V_T ; whereas reverse back biasing (RBB) refers to biasing that increases V_T .

[0033] For purposes of illustration, FIG. 2B shows PFETs **210**, **220**, **230**, **240** on insulator layer **205** over a Pwell **203** within semiconductor substrate **201**. Thus, in this example, PFETs **210**, **220**, **230**, **240** are either LVT or SLVT PFETs. A buried Nwell **202** can be below Pwell **203** and additional Nwells **204** can extend from the top surface of semiconductor substrate **201** to buried Nwell **202** and can laterally surround Pwell **203**, thereby isolating Pwell **203**. In this case, adjacent portions of insulator layer **205** and Pwell **203** aligned below each of channel regions **215**, **225**, **235**, **245** effectively form back gates **214**, **224**, **234**, **244** for PFETs **210**, **220**, **230**, and **240**. An isolation region **209** (e.g., a shallow trench isolation (STI) region) can extend through semiconductor layer **206** and laterally surrounds that portion of semiconductor layer **206** containing the active device regions for the PFETs **210**, **220**, **230**, and **240**. Additionally, a well tap opening can extend vertically through isolation region **209** and insulator layer **205** to Pwell **203**. A Pwell tap **207** can be immediately adjacent to Pwell **203** in the well tap opening and can have P+ conductivity. Pwell tap **207** can be an in situ-doped epitaxial semiconductor layer immediately adjacent to the top surface of semiconductor substrate **201** at Pwell **203**. Alternatively, Pwell tap **207** can be a shallow P+doped region within Pwell **203**. Optionally, Pwell **203** can be biased (e.g., via well tap **207**) to concurrently adjust the V_T s of the PFETs **210**, **220**, **230**, and **240**. For example, Pwell tap **207** can be electrically connected (e.g., via middle of the line (MOL) and back end of the line (BEOL) interconnects) to first power rail **291**. Thus, back gates **214**, **224**, **234**, **244** are concurrently back biased using V_1 , thereby reducing the V_T s of the PFETs.

[0034] In operation, pre-driver **120** receives Pin **118** and Pinb **119** at first pre-driver input node **251** and second pre-driver input node **252**, respectively. When Pin **118** is at V_2 and Pinb **119** is at GND, first PFET **210** turns off and second PFET **220** turns on. As a result of second PFET **220** turning on, the voltage level at second pre-driver output node **262** is pulled down to V_1 (i.e., Dinb **129** drops to V_1). Additionally, since V_1 at second pre-driver output node **262** is also applied to third gate **233** of third PFET **230** (because of cross-coupling), third PFET **230** turns on and the voltage level on first pre-driver output node **261** is pulled-up to V_2 . Thus, Din **128** rises to V_2 . Additionally, since V_2 at first pre-driver output node **261** is applied to fourth gate **243** of fourth PFET **240** (again because of cross-coupling), fourth PFET **240** turns off and the voltage level on the second pre-driver output node **262** remains low (i.e., at V_1). When Pin **118** switches to GND and Pinb **119** switches to V_2 , second PFET **220** turns off and first PFET **210** turns on. As a result, the voltage level at first pre-driver output node **261** is pulled down to V_1 . Thus, Din **128** drops to V_1 . Additionally, since V_1 at first pre-driver output node **261** is applied to fourth gate **243** of fourth PFET **240** (because of cross-coupling), fourth PFET **240** turns on and the voltage level on the second pre-driver output node **262** is pulled-up to V_2 . Thus, Dinb **129** rises to V_2 . Additionally, since V_2 at the second pre-driver output node **262** is applied to third gate **233** of third PFET **230**, third PFET **230** turns off and the voltage level on the first pre-driver output node **261** remains low (i.e., at V_1). Thus, at no point are V_{GD} , V_{GS} , or V_{DS} of any of PFETs **210**, **220**, **230** or **240** greater than the voltage rating thereof

or, more particularly, greater than an amount equal to the voltage rating plus some allowable amount over the voltage rating (e.g., $1.8V+0.18V=1.98V$ maximum limit). Furthermore, the pre-driver **120** exhibits improved performance (e.g., reduced delay, reduced static and short circuit currents, etc.) and reduced PVT variations as compared to pre-drivers with a combination of NFETs and PFETs and/or with a combination of low and high voltage devices.

[0035] As mentioned above, the disclosed compact high-voltage pre-driver **120** can be incorporated into a multi-stage wordline driver circuit. **100** between a voltage level shifter **110** and a driver **130**.

[0036] Voltage level shifter **110** can be connected to receive a low voltage wordline select signal (WLsel-lv) **108**, which switches between V1 and GND. Voltage level shifter **110** can further be configured to output Pin **118** (which, as mentioned above, switches between V2 and GND) and Pinb **119** (which, as mentioned above, switches between GND and V2). Voltage level shifter **110** can, for example, be connected between a ground (GND) rail **291** and second power rail **292** (which is at V2). FIG. 3 is a schematic diagram illustrating one example of such a voltage level shifter **110** that could be included in the multi-stage wordline driver circuit **100**. In this example, voltage level shifter **110** can include four FETs including: NFETs **310** and **320** and PFETs **330** and **340**. NFET **310** and PFET **330** can be series-connected between GND **391** and second power rail **292**. Similarly, NFET **320** and PFET **340** can be series-connected between GND **391** and second power rail **292**. Additionally, PFETs **330** and **340** can be cross-coupled. Voltage level shifter **110** can further include first and second voltage level shifter input nodes **351** and **352** electrically connected to the gates of NFETs **310** and **320**, respectively. Voltage level shifter input node **351** can be electrically to receive a low voltage wordline select signal (WLsel-lv) **108**. WLsel-lv **108** can switch between V1 and GND. Voltage level shifter input node **351** can be electrically connected to receive an inverted low voltage wordline select signal (WLsel-lvb) **109**. WLsel-lvb **109** can be inverted with respect to WLsel-lv **108** such that, when WLsel-lv **108** switches from V1 to GND, WLsel-lvb **109** switches from GND to V1 and vice versa. Voltage level shifter **110** can further include first and second voltage level shifter output nodes **361** and **362**. First voltage level shifter output node **361** can be electrically connected to an interconnect between drain regions of NFET **310** and PFET **330** and can output Pinb **119**. Second voltage level shifter output node **362** can be electrically connected to an interconnect between drain regions of NFET **320** and PFET **340** and can output Pin **118**.

[0037] In operation, voltage level shifter **110** receives WLsel-lv **108** and WLsel-lvb **109** at first voltage level shifter input node **351** and second voltage level shifter input node **352**, respectively. When WLsel-lv **108** is at V1 and WLsel-lvb **109** is at GND, NFET **310** turns on and NFET **320** turns off. As a result of NFET **310** turning on, the voltage level at first voltage level shifter output node **361** is pulled down to GND (i.e., Pinb **119** drops to GND). Additionally, since GND at first voltage level shifter output node **361** is also applied to the gate of PFET **340** (because of cross-coupling), PFET **340** turns on and the voltage level on second voltage level shifter output node **362** is pulled-up to V2. Thus, Pin **118** rises to V2. Additionally, since V2 at second voltage level shifter output node **362** is applied to the gate of PFET **330** (again because of cross-coupling), PFET **330** turns off and the voltage level on first voltage level shifter output node **361** remains low (i.e., at GND). When WLsel-lv **108** switches to GND and WLsel-lvb **109** switches to V1, NFET **310** turns off and NFET **320** turns on. As a result, the voltage level at second voltage level shifter output node **362** is pulled down to GND. Thus, Pin **118** drops to GND. Additionally, since GND at second voltage level shifter output node **362** is applied to gate of PFET **330** (because of cross-coupling), PFET **330** turns on and the voltage level on the first voltage level shifter output node **361** is pulled-up to V2. Thus, Pinb **119** rises to V2. Additionally, since V2 at the first voltage level shifter output node **361** is applied to the gate of PFET **340**, PFET **340** turns off and the voltage level on the second voltage level shifter output node **362** remains low (i.e., at GND).

[0038] Driver **130** can be connected to receive Din **128** from pre-driver **120**. Driver **120** can further be configured to output a high voltage wordline select signal (WLsel-hv) **138** that, as mentioned

above, switches between V2 and GND. Driver **130** can, for example, be connected between GND rail **391** and a variable power rail **492**. As discussed in greater detail below, variable power rail **492** receives a variable wordline supply voltage (vVwls) **539** from a variable power supply circuit **500** and vVwls **539** that switches between V1 and V2 concurrent with Din switching from V2 to V1 and vice versa.

[0039] FIG. **4** is a schematic diagram illustrating one example of such a driver **130** that could be included in the multi-stage wordline driver circuit **100**. In this example, driver **130** can include four stacked FETs including: two NFETs **410** and **420** and two PFETs **430** and **440** series-connected between GND **391** and variable power rail **492**. Gate of PFET **440** can receive Din **128** (which switches between V2 and V1) as an input. Gates of PFET **430** and NFET **420** can be continuously biased with V1 (e.g., tied to the first power rail). The gate of NFET **410** can be connected to receive WLsel-lv **108** (which, as mentioned above, switches between V1 and GND). Driver **130** can further have a driver output node **461** connected to an interconnect between NFET **420** and PFET **430**. Driver output node **461** can output WLsel-hv **138** (which, as mentioned above, switches between V2 and GND).

[0040] As mentioned above, vVwls **539** on variable power rail **492** is supplied by a variable power supply circuit **500**. FIG. **5** is a schematic diagram illustrating an example of such a variable power supply circuit **500**. In this example, variable power supply circuit **500** is a multi-stage variable power supply circuit and, like the multi-stage wordline driver circuit **100** of FIG. **1**, it may include a compact high voltage pre-driver **520**, as disclosed herein. That is, this multi-stage variable power supply circuit **500** includes a voltage level shifter **510**, a buffer **530**, and a novel compact high voltage pre-driver **520** connected to voltage level shifter **510** and to buffer **530**. Multi-stage variable power supply circuit **500** and components thereof can be implemented in an advanced semiconductor-on-insulator processing technology node (e.g., a fully depleted semiconductor-on-insulator processing technology node, such as a fully depleted silicon-on-insulator (FDSOI) processing technology node).

[0041] It should be noted that the same voltage level shifter **110/510** can be shared by multi-stage wordline driver circuit **100** and multi-stage variable power supply circuit **500**. In this case, the same Pin **118** (which switches between V2 and GND) and the same Pinb **119** (which switches between GND and V2) will be output from the voltage level shifter to both pre-driver **120** of multi-stage wordline driver circuit **100** and pre-driver **520** of multi-stage variable power supply circuit **500**. Alternatively, discrete voltage level shifters **110** and **510** can be employed in circuits **100** and **500**. In this case, voltage level shifters **110**, **510** can have the same configuration (e.g., as described above and illustrated in FIG. **3**), can have the same input signals WLsel-lv and WLsel-lvb (also as described above) and, in response to those input signals, can each output separate Pin and Pinb signals. Pre-driver **520** of multi-stage variable power supply circuit **500** can have the same configuration as pre-driver **120** of multi-stage wordline driver circuit **100**, described in detail above and illustrated in FIG. **2A**). Specifically, FIG. **6** is a schematic diagram illustrating pre-driver **520** and, like pre-driver **120**, pre-driver **520** is electrically connected between first power rail **291** (which as mentioned above is at the relatively low first voltage (V1)) and second power rail **292** (which as mentioned above is at the relatively high second voltage (V2)).

[0042] Pre-driver **520** includes four PFETs **610**, **620**, **630**, **640** and, particularly, a first PFET **610**, a second PFET **620**, a third PFET **630** and a fourth PFET **640**. Each PFET **610**, **620**, **630**, and **640** is a low voltage symmetric PFET. Additionally, PFETs **610**, **620**, **630**, and **640** can have the same relatively low voltage ratings for the maximum operating VDS, the maximum operating VGS, and the maximum operating VGD. First PFET **610** and third PFET **630** are series-connected between first power rail **291** and second power rail **292**. Similarly, second PFET **620** and fourth PFET **640** are series-connected between first power rail **291** and second power rail **292**. Additionally, third PFET **630** and fourth PFET **640** are cross-coupled. That is, the gate of third PFET **630** is electrically connected to the drain region of fourth PFET **640** and the gate of fourth PFET **640** is

electrically connected to the drain region of third PFET **630**.

[0043] Pre-driver **520** further includes pre-driver input nodes **651** and **652** electrically connected to receive Pin and Pinb and to apply Pin and Pinb to gates of the first and second PFETs **610** and **620**, respectively. As mentioned above, Pin switches between V2 and GND and Pinb is inverted with respect to Pin such that, when Pin switches from V2 to GND, Pinb switches from GND to V2 and vice versa. Pre-driver **520** further includes pre-driver output nodes **661** and **662** at junctions between first PFET **610** and third PFET **630** and between second PFET **620** and fourth PFET **640**, respectively. Pre-driver output node **661** outputs a pre-driver output signal (Pout) that switches between V2 and V1. Pre-driver output node **662** outputs an inverted pre-driver output signal (Poutb). In the case of the variable power supply circuit **500**, Poutb can also be referred to as buffer input signal (Bin) **529**. Bin **529** can be inverted with respect to Pout such that, when Pout switches from V2 to V1, Bin **529** switches from V1 to V2 and vice versa.

[0044] Operation of pre-driver **520** within multi-stage variable power supply circuit **500** can be essentially the same as that described in detail above with regard to operation of pre-driver **120** within multi-stage wordline driver circuit **100**. Thus, within pre-driver **520**, none of the low voltage PFETs operate outside the SOA (i.e., voltage ratings are not violated) and pre-driver **520** similarly exhibits improved performance (e.g., reduced delay, reduced static and short circuit currents, etc.) and reduced PVT variations.

[0045] Buffer **530** can receive Bin **529** and can output vVwls **539** to the variable power rail **492**. vVwls **539**, like Bin **529**, can switch between V1 and V2 concurrent with Din switching from V2 to V1 and vice versa. As with conventional buffers, buffer **530** effectively connects a high-impedance source with a low-impedance load without significant attenuation or signal distortion. Referring again to FIG. **4** and, particularly, to the detailed schematic drawing of driver **130** of multi-stage wordline driver circuit **100**.

[0046] In operation, when driver **130** is selected, vVwls **539** on variable power rail **492** will switch to V2 and Din on the gate of PFET **440** will switch to V1. Thus, VSG of PFET **440** will be $V2 - V1 > V_{TH}$ such that PFET **440** turns on. As a result of PFET **440** turning on, the voltage level at the drain of PFET **430** and thereby at the source of PFET **430** will be pulled up to V2. With V1 applied to the gate of PFET **430**, VSG of PFET **430** will be $V2 - V1 > V_{TH}$ such that PFET **430** also turns on. As a result of PFETs **440** and **430** turning on, the voltage level at driver output node **461** will be pulled up to V2 such that WLsel-hv **138** is at V2. Concurrently, WLsel-lv **108** applied to the gate of NFET **410** will be at GND. Thus, VGS of NFET **410** will be at $GND - GND = 0V < V_{TH}$ such that NFET **410** turns off. As a result of NFET **410** turning off, the drain of NFET **410** will be floating. With V1 applied to the gate of NFET **420**, the source of NFET **420** and the drain of NFET **410** will settle at V1 (i.e., at the gate voltage of NFET **420**).

[0047] When driver **130** is unselected, vVwls **539** on variable power rail **492** could be at either V2 or V1.

[0048] Specifically, when driver **130** is unselected, vVwls **539** is at V2 and Din **128** on the gate of PFET **440** is at V2, VSG of PFET **440** will be at $V2 - V2 = 0V < V_{TH}$ such that PFET **440** turns off. As a result of PFET **440** turning off, the drain of PFET **440** will be floating. With V1 applied to the gate of PFET **430**, the source of PFET **430** and the drain of PFET **440** will settle at V1 (i.e., at the gate voltage of PFET **430**). Concurrently, WLsel-lv **108** applied to the gate of NFET **410** will be at V1. Thus, VGS of NFET **410** will be $V1 - GND > V_{TH}$ such that NFET **410** turns on. As a result of NFET **410** turning on, the voltage at the drain of NMOS **410** will be pulled down to GND. With V1 applied to the gate of NFET **420**, VGS of NFET **420** will be $V1 - GND > V_{TH}$, such that NFET **420** turns on. As a result of NFETs **410** and **420** turning on, the voltage level at driver output node **461** is pulled down to GND.

[0049] When driver **130** is unselected, vVwls **539** is at V1, and Din **128** on the gate of PFET **440** is at V2, VSG of PFET **440** is $V1 - V2 < V_{TH}$ such that PFET **440** turns off. As a result of PFET **440** turning off, the drain of PFET **440** will be floating. With V1 applied to the gate of PFET **430**, the

source of PFET **430** and drain of PFET **440** will settle at **V1** (i.e., at the gate voltage of PFET **430**). Concurrently, WLsel-lv **108** applied to the gate of NFET **410** will be at **V1**. Thus, VGS of NFET **410** will be $V1 - GND > V_{TH}$ such that NFET **410** turns on. As a result of NFET **410** turning on, the voltage at the drain of NFET **410** is pulled down to GND. With **V1** applied to the gate of NFET **420**, VGS of NFET **420** will be at $V1 - GND > V_{TH}$ such that NFET **420** turns on. As a result of NFETs **410** and **420** turning on, the voltage level at driver output node **461** is pulled down to GND. [0050] When driver **130** is unselected, vV_{wls} **539** is at **V1**, and Din **128** on the gate of PFET **440** is at **V1**, VSG of PFET **440** is $V1 - V1 = 0V < V_{TH}$ such that PFET **440** turns off. As a result of PFET **440** turning off, the drain of PFET **440** will be floating. With **V1** applied to the gate of PFET **430**, the source of PFET **430** and the drain of PFET **440** will settle at **V1** (i.e., at the gate voltage of PFET **430**). Concurrently, WLsel-lv **108** applied to the gate of NFET **410** will be at **V1**. Thus, VGS of NFET **410** will be $V1 - GND > V_{TH}$ such that NFET **410** turns on. As a result of NFET **410** turning on, the voltage at the drain of NFET **410** will be pulled down to GND. With **V1** applied to the gate of NFET **420**, VGS of NFET **420** will be $V1 - GND > V_{TH}$ such that NFET **420** turns on. Again, as a result of NFETs **410** and **420** turning on, the voltage level at driver output node **461** is pulled down to GND.

[0051] The voltage level shifter and driver examples provided in FIGS. **3** and **4** and described above are provided for illustration purposes and are not intended to be limiting. Alternatively, any other suitable voltage level shifter **110** and/or driver **130** could be included in the multi-stage wordline driver circuit **100** of FIG. **1** in addition to pre-driver **120** (as shown in FIG. **2A**).

[0052] FIG. **7** is a graph including curves **701**, **702** and **703** representing Pout, Poutb, and output current, respectively, over time for the disclosed compact high-voltage pre-driver (e.g., pre-driver **120** or pre-driver **520**). As illustrated, **V1** of Pout and Poutb are at approximately 1.8 with only minimal PVT variations (i.e., **V1** may range from 1.62V-1.98V). Furthermore, the delay between switching of Pout and switching of Poutb is relatively small and reduced as compared to prior art pre-drivers. For example, delay (D) between Pout and Poutb in the disclosed pre-driver can be less than 350 picoseconds (ps), such as at 321 ps, whereas D between Pout and Poutb of prior art pre-drivers is typically over 550 ps (e.g., 566 ps). Also, static current (I_s) and peak short circuit current (I_{sc}) (i.e., the maximum possible instantaneous value of a short circuit current) of Iout are relatively small and reduced as compared to prior art pre-drivers. For example, as illustrated, I_s is approximately 3.26 nA and I_{sc} is approximately 2.2 mA, whereas I_s and I_{sc} in prior art pre-drivers are typically over 15 nA (e.g., 15.7 nA) and over 13 mA (e.g., 13.3 mA), respectively. Thus, the disclosed compact high-voltage pre-driver exhibits reduced power consumption as compared to prior art pre-drivers. It should be noted that the multi-stage wordline driver circuit **100** of FIG. **1** and the multi-stage variable power supply circuit **500** of FIG. **5** are just two examples of circuits that can incorporate a compact high voltage pre-driver as disclosed herein. Alternatively, any other circuit requiring a pre-driver could similarly incorporate the disclosed compact high-voltage pre-driver.

[0053] In the method and structures described above, a semiconductor material refers to a material whose conducting properties can be altered by doping with an impurity. Exemplary semiconductor materials include, for example, silicon-based semiconductor materials (e.g., silicon, silicon germanium, silicon germanium carbide, silicon carbide, etc.) and III-V compound semiconductors (i.e., compounds obtained by combining group III elements, such as aluminum (Al), gallium (Ga), or indium (In), with group V elements, such as nitrogen (N), phosphorous (P), arsenic (As) or antimony (Sb)) (e.g., GaN, InP, GaAs, or GaP). A pure semiconductor material and, more particularly, a semiconductor material that is not doped with an impurity for the purposes of increasing conductivity (i.e., an undoped semiconductor material) is referred to in the art as an intrinsic semiconductor. A semiconductor material that is doped with an impurity for the purposes of increasing conductivity (i.e., a doped semiconductor material) is referred to in the art as an extrinsic semiconductor and will be more conductive than an intrinsic semiconductor made of the

same base material. That is, extrinsic silicon will be more conductive than intrinsic silicon; extrinsic silicon germanium will be more conductive than intrinsic silicon germanium; and so on. Furthermore, it should be understood that different impurities (i.e., different dopants) can be used to achieve different conductivity types (e.g., P-type conductivity and N-type conductivity) and that the dopants may vary depending upon the different semiconductor materials used. For example, a silicon-based semiconductor material (e.g., silicon, silicon germanium, etc.) is typically doped with a Group III dopant, such as boron (B) or indium (In), to achieve P-type conductivity, whereas a silicon-based semiconductor material is typically doped with a Group V dopant, such as arsenic (As), phosphorous (P) or antimony (Sb), to achieve N-type conductivity. A gallium nitride (GaN)-based semiconductor material is typically doped with magnesium (Mg) to achieve P-type conductivity and with silicon (Si) or oxygen to achieve N-type conductivity. Those skilled in the art will also recognize that different conductivity levels will depend upon the relative concentration levels of the dopant(s) in a given semiconductor region.

[0054] It should be understood that the terminology used herein is for the purpose of describing the disclosed structures and methods and is not intended to be limiting. For example, as used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, as used herein, the terms “comprises,” “comprising,” “includes,” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Furthermore, as used herein, terms such as “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “under,” “below,” “underlying,” “over,” “overlying,” “parallel,” “perpendicular,” etc., are intended to describe relative locations as they are oriented and illustrated in the drawings (unless otherwise indicated) and terms such as “touching,” “in direct contact,” “abutting,” “directly adjacent to,” “immediately adjacent to,” etc., are intended to indicate that at least one element physically contacts another element (without other elements separating the described elements). The term “laterally” is used herein to describe the relative locations of elements and, more particularly, to indicate that an element is positioned to the side of another element as opposed to above or below the other element, as those elements are oriented and illustrated in the drawings. For example, an element that is positioned laterally adjacent to another element will be beside the other element, an element that is positioned laterally immediately adjacent to another element will be directly beside the other element, and an element that laterally surrounds another element will be adjacent to and border the outer sidewalls of the other element. The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed.

[0055] The method as described above is used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip is mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other higher level carrier) or in a multichip package (such as a ceramic carrier that has either or both surface interconnections or buried interconnections). In any case the chip is then integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either (a) an intermediate product, such as a motherboard, or (b) an end product. The end product can be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

[0056] The descriptions of the various disclosed embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limiting. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of

the disclosed embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a first input node; a second input node; and four transistors including: a first P-channel field effect transistor (PFET) with a first gate connected to the first input node; a second PFET with a second gate connected to the second input node; a third PFET; and a fourth PFET, wherein the first PFET and the third PFET are connected in series between a first power rail at a first voltage and a second power rail at a second voltage greater than the first voltage, wherein the second PFET and the fourth PFET are connected in series between the first power rail and the second power rail, wherein the third PFET and the fourth PFET are cross-coupled, wherein the first input node is connected to receive a pre-driver input signal that switches between the second voltage and ground, and wherein the second input node is connected to receive an inverted pre-driver input signal.
2. The structure of claim 1, further comprising an output node at a junction between the first PFET and the third PFET, wherein the output node outputs a pre-driver output signal that switches between the second voltage and the first voltage.
3. The structure of claim 1, wherein the four transistors have identical voltage ratings for an operating gate-source voltage, an operating gate-drain voltage, and an operating source-drain voltage.
4. The structure of claim 3, wherein the first voltage is set based on the voltage ratings.
5. The structure of claim 4, wherein the second voltage is twice the first voltage.
6. The structure of claim 4, wherein the first voltage is 1.8V and the second voltage is 3.6V.
7. The structure of claim 1, wherein the first PFET and the second PFET have a first size and wherein the third PFET and the fourth PFET have a second size that is smaller than the first size.
8. The structure of claim 1, wherein the four transistors have back gates including adjacent portions of an insulator layer and a well region in a semiconductor substrate below the insulator layer and wherein the well region is biased at the first voltage.
9. A structure comprising: a voltage level shifter connected to receive a voltage level shifter input signal that switches between a first voltage and ground, wherein the voltage level shifter outputs a pre-driver input signal that switches between a second voltage that is greater than the first voltage and ground and further outputs an inverted pre-driver input signal; a pre-driver including: a first input node; a second input node; four transistors including: a first P-channel field effect transistor (PFET) with a first gate connected to the first input node; a second PFET with a second gate connected to the second input node; a third PFET; and a fourth PFET, wherein the first PFET and the third PFET are connected in series between a first power rail at a first voltage and a second power rail at a second voltage greater than the first voltage, wherein the second PFET and the fourth PFET are connected in series between the first power rail and the second power rail, wherein the third PFET and the fourth PFET are cross-coupled, wherein the first input node is connected to receive the pre-driver input signal, and wherein the second input node is connected to receive the inverted pre-driver input signal; an output node at a junction between the first PFET and the third PFET, wherein the output node outputs a pre-driver output signal that switches between the second voltage and the first voltage; and a device connected to receive the pre-driver output signal.
10. The structure of claim 9, wherein the device is a driver and wherein the driver outputs a wordline select signal that switches between the second voltage and ground.
11. The structure of claim 9, wherein the device is a buffer.
12. The structure of claim 9, wherein the four transistors have identical voltage ratings for an

operating gate-source voltage, an operating gate-drain voltage, and an operating source-drain voltage and the first voltage is set based on the voltage ratings.

13. The structure of claim 9, wherein the second voltage is twice the first voltage.

14. The structure of claim 9, wherein the first PFET and the second PFET have a first size, and wherein the third PFET and the fourth PFET have a second size that is smaller than the first size.

15. The structure of claim 9, wherein the four transistors have back gates adjacent portions of an insulator layer and a well region in a substrate below the insulator layer and wherein the well region is biased at the first voltage.

16. A structure comprising: a voltage level shifter connected to receive a first wordline select signal that switches between a first voltage and ground, wherein the voltage level shifter outputs a pre-driver input signal that switches between a second voltage that is greater than the first voltage and ground and further outputs an inverted pre-driver input signal; a pre-driver including: a first input node; a second input node; four transistors including: a first P-channel field effect transistor (PFET) with a first gate connected to the first input node; a second PFET with a second gate connected to the second input node; a third PFET; and a fourth PFET, wherein the first PFET and the third PFET are connected in series between a first power rail at the first voltage and a second power rail at the second voltage, wherein the second PFET and the fourth PFET are connected in series between the first power rail and the second power rail, wherein the third PFET and the fourth PFET are cross-coupled, and wherein the first input node is connected to receive the pre-driver input signal, and wherein the second input node is connected to receive the inverted pre-driver input signal; an output node at a junction between the first PFET and the third PFET, wherein the output node outputs a driver input signal that switches between the second voltage and the first voltage; and a driver connected to receive the driver input signal, wherein the driver outputs a second wordline select signal that switches between the second voltage and ground.

17. The structure of claim 16, wherein the four transistors have identical voltage ratings for an operating gate-source voltage, an operating gate-drain voltage, and an operating source-drain voltage and wherein the first voltage is set based on the voltage ratings.

18. The structure of claim 16, wherein the second voltage is twice the first voltage.

19. The structure of claim 16, wherein the first PFET and the second PFET have a first size and wherein the third PFET and the fourth PFET have a second size that is smaller than the first size.

20. The structure of claim 16, wherein the four transistors have back gates including portions of an insulator layer and of a well region in a semiconductor substrate below the insulator layer and wherein the well region is biased at the first voltage.
